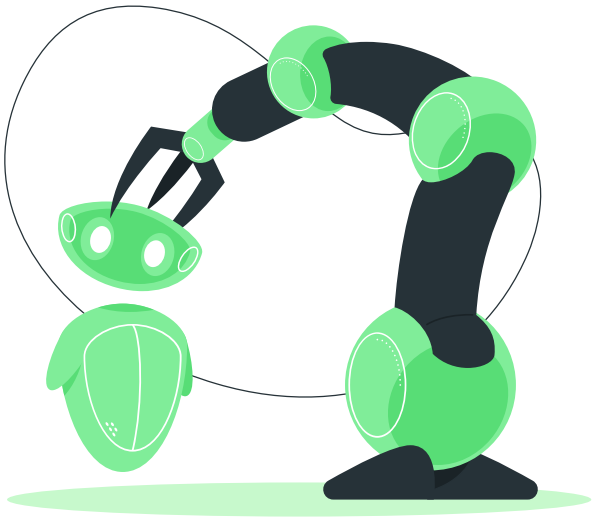
Robotics
CS 3015

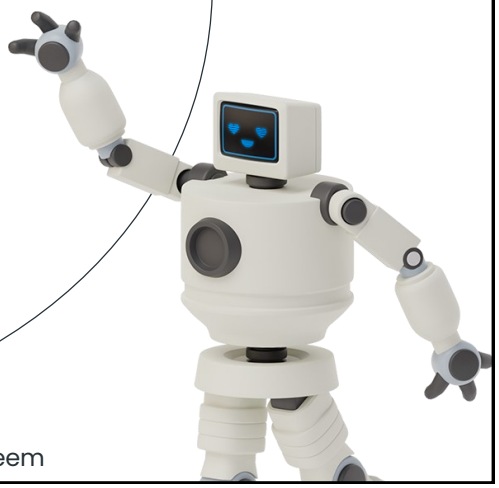
Umarah Qaseem



1

7b

- Inverse Kinematics
- Analytic Solution
- Tangent complex cases
- Spherical Wrist
- Numeric Solution
- Comparison, Pros and cons
- Practical Issues
- Redundant Robots



2

Kinematics & Dynamics



Robotics – Umarah Qaseem

8

Kinematics and Dynamics

- **Kinematics** is the branch of mechanics that studies the motion of bodies without considering the forces that cause them to move. Often referred to as the "Geometry of motion".
- **Dynamics** is the branch of mechanics concerned with the motion of bodies under the action of forces.

Robotics – Umarah Qaseem

9

Robot Kinematics

- Two kinematic problems in Robotics

- **Forward Kinematics**: Given joint angles, we can find the robot's tool tip.
- **Inverse Kinematics**: Given the pose of robot tool tip, we find the required joint angles.

10



Inverse Kinematics

11

Inverse Kinematics

- What joint angles to set to achieve a certain end-effector pose.

$$\xi_N = \mathcal{K}(\boldsymbol{q})$$

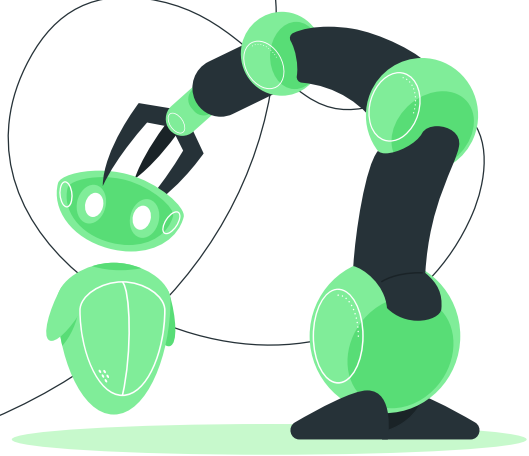
$$\boldsymbol{q} = \{q_j, j \in [1 \cdots N]\}$$

$$\boldsymbol{q} = \mathcal{K}^{-1}(\xi_N)$$

Robotics – Umarah Qaseem

12

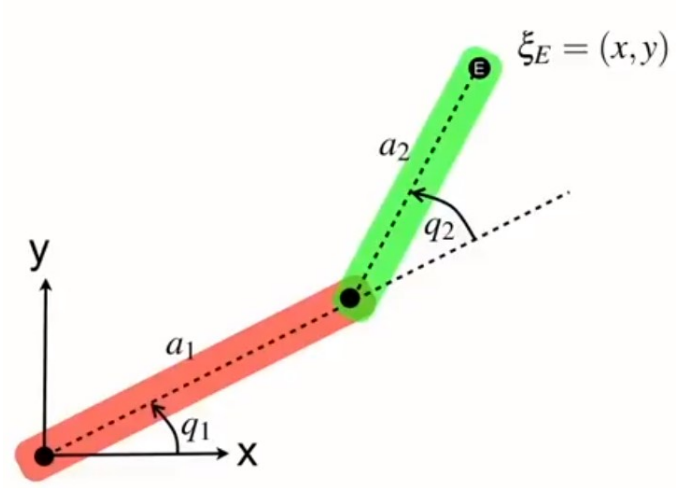
Inverse Kinematics Geometric Approach



Robotics – Umarah Qaseem

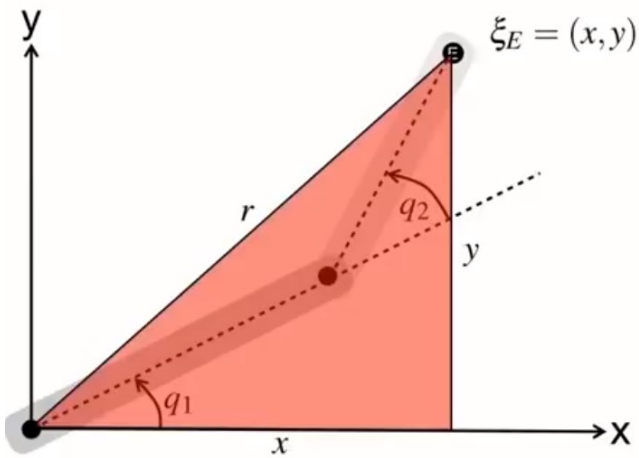
13

IK for 2 Joint Arm – Geometric Approach



14

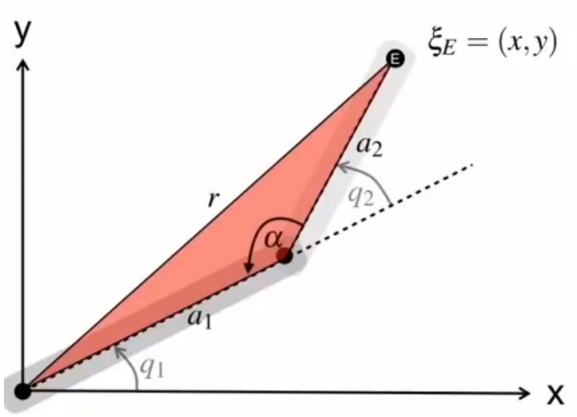
IK or 2 joint arm – Geometric Approach



$$r^2 = x^2 + y^2$$

15

IK or 2 joint arm – Geometric Approach



***We need to find angle alpha so we apply cosine rule**

Cosine Rule

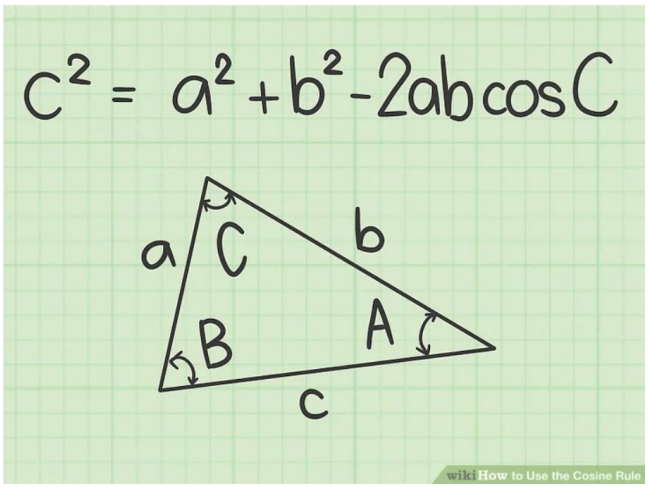
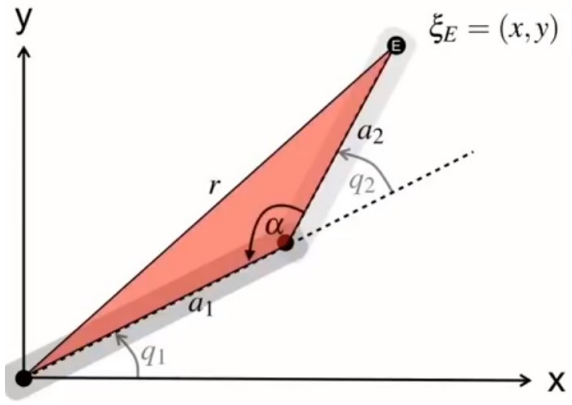


Image Credits: wikiHow

IK for 2 joint arm – Geometric Approach



$$r^2 = x^2 + y^2$$

$$r^2 = a_1^2 + a_2^2 - 2a_1 a_2 \cos\alpha$$

$$\cos\alpha = \frac{a_1^2 + a_2^2 - r^2}{2a_1 a_2}$$

$$= \frac{a_1^2 + a_2^2 - x^2 - y^2}{2a_1 a_2}$$

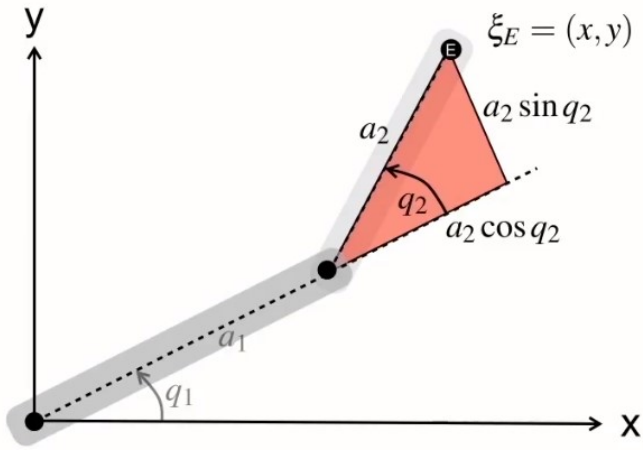
$$q_2 = \pi - \alpha$$

$$\cos q_2 = -\cos\alpha$$

$$\cos q_2 = \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1 a_2}$$

18

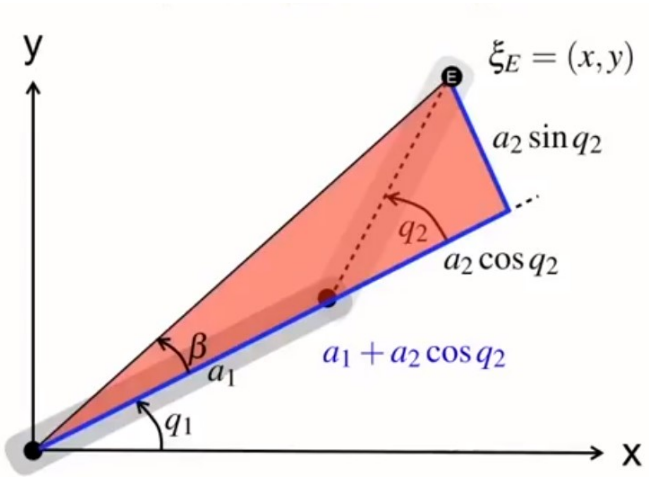
IK for 2 joint arm – Geometric Approach



$$\sin q_2 = \sqrt{1 - \cos q_2^2}$$

19

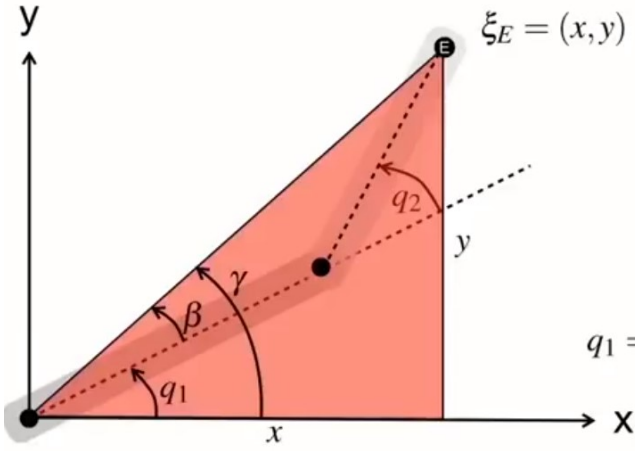
IK for 2 joint arm – Geometric Approach



$$\beta = \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

20

IK for 2 joint arm – Geometric Approach



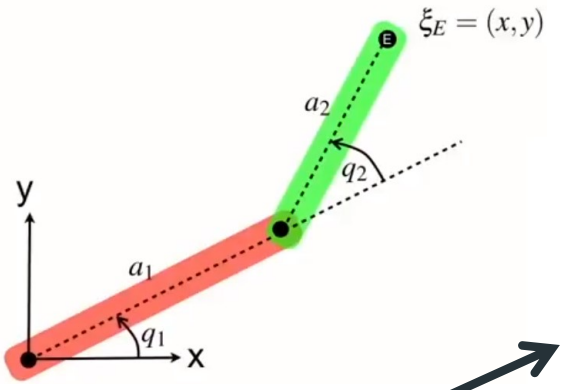
$$\gamma = \tan^{-1} \frac{y}{x}$$

$$q_1 = \gamma - \beta$$

$$q_1 = \tan^{-1} \frac{y}{x} - \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

21

IK for 2 joint arm – Geometric Approach



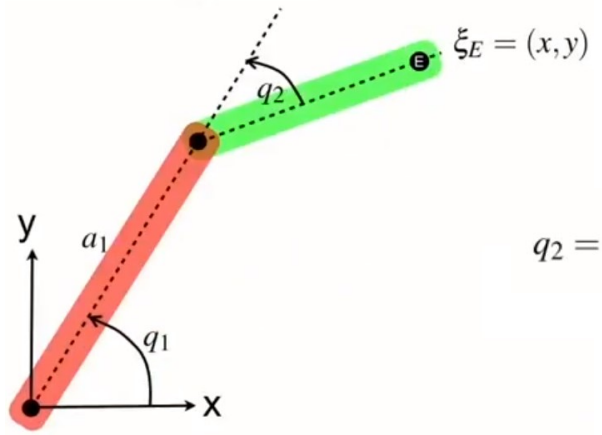
Inverse Kinematic Solution

$$\cos q_2 = \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

$$q_2 = \cos^{-1} \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

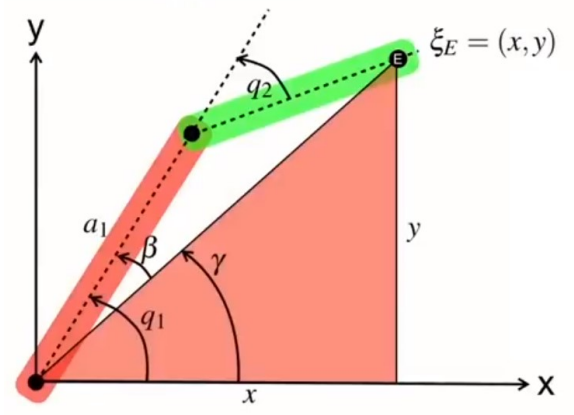
$$q_1 = \tan^{-1} \frac{y}{x} - \tan^{-1} \frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

IK for 2 joint arm – Geometric Approach



$$q_2 = -\cos^{-1} \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1a_2}$$

IK for 2 joint arm – Geometric Approach

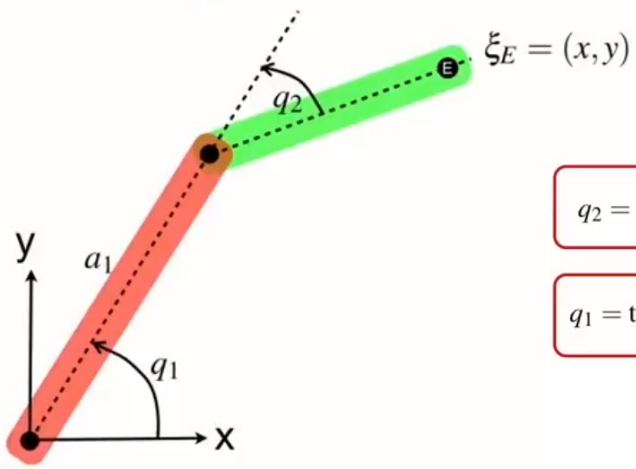


$$\gamma = \tan^{-1}\frac{y}{x}$$

$$q_1 = \gamma + \beta$$

$$q_1 = \tan^{-1}\frac{y}{x} + \tan^{-1}\frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

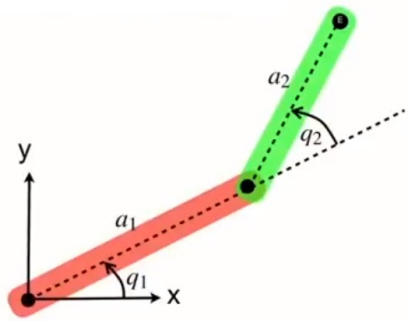
IK for 2 joint arm – Geometric Approach



$$q_2 = -\cos^{-1}\frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1 a_2}$$

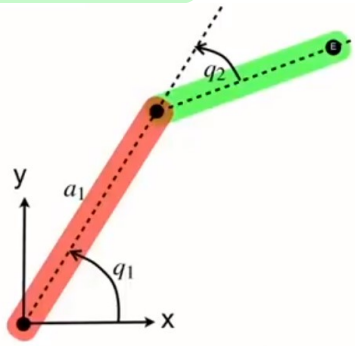
$$q_1 = \tan^{-1}\frac{y}{x} + \tan^{-1}\frac{a_2 \sin q_2}{a_1 + a_2 \cos q_2}$$

IK of 2 joint arm – Geometric Approach



$$q_2 = \cos^{-1}\frac{x^2+y^2-a_1^2-a_2^2}{2a_1a_2}$$

$$q_1 = \tan^{-1}\frac{y}{x} - \tan^{-1}\frac{a_2\sin q_2}{a_1+a_2\cos q_2}$$

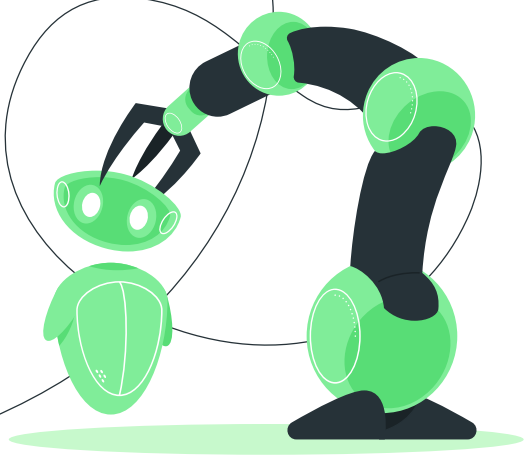


$$q_2 = -\cos^{-1}\frac{x^2+y^2-a_1^2-a_2^2}{2a_1a_2}$$

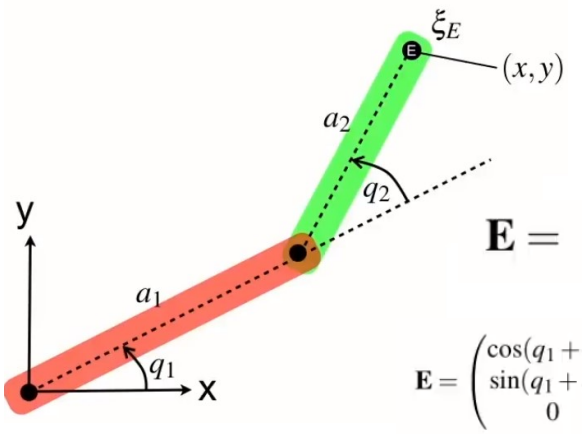
$$q_1 = \tan^{-1}\frac{y}{x} + \tan^{-1}\frac{a_2\sin q_2}{a_1+a_2\cos q_2}$$

Inverse Kinematics Analytic Approach

Algebraic



Tool Tip pose



$$\mathbf{E} = \mathbf{R}(q_1)\,\mathbf{T}_x(a_1)\,\mathbf{R}(q_2)\,\mathbf{T}_x(a_2)$$

$$\mathbf{E} = \begin{pmatrix} \cos(q_1+q_2) & -\sin(q_1+q_2) & a_2\cos(q_1+q_2)+a_1\cos q_1 \\ \sin(q_1+q_2) & \cos(q_1+q_2) & a_2\sin(q_1+q_2)+a_1\sin q_1 \\ 0 & 0 & 1 \end{pmatrix}$$

28

Tool Tip pose

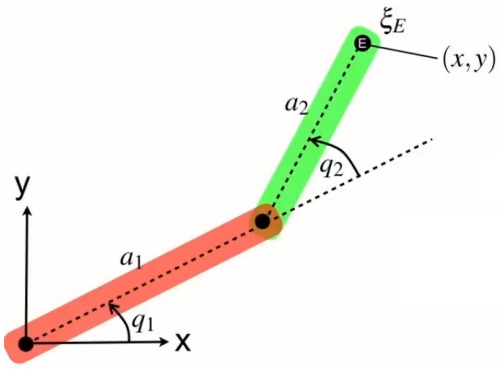
$$\mathbf{E} = \begin{pmatrix} \cos(q_1+q_2) & -\sin(q_1+q_2) & a_2\cos(q_1+q_2)+a_1\cos q_1 \\ \sin(q_1+q_2) & \cos(q_1+q_2) & a_2\sin(q_1+q_2)+a_1\sin q_1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_2\cos(q_1+q_2)+a_1\cos q_1 \\ a_2\sin(q_1+q_2)+a_1\sin q_1 \end{pmatrix}$$

29

Solve!

- Square and Add

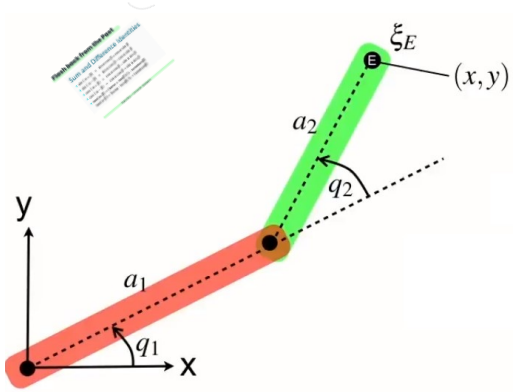


$$x = a_2 \cos(q_1 + q_2) + a_1 \cos q_1$$
$$y = a_2 \sin(q_1 + q_2) + a_1 \sin q_1$$

$$x^2 + y^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos q_2$$

$$\cos q_2 = \frac{x^2 + y^2 - a_1^2 - a_2^2}{2a_1 a_2}$$

Analytic/Algebraic Solution



$$x = a_2 \cos(q_1 + q_2) + a_1 \cos q_1$$
$$y = a_2 \sin(q_1 + q_2) + a_1 \sin q_1$$

After applying sum of a angle identities and substituting cosq2 by C2 and sinq2 by S2

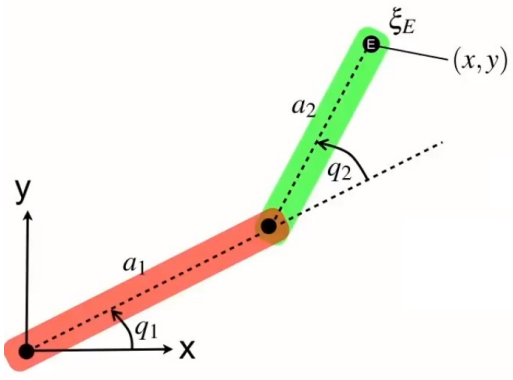
$$x = (a_1 + a_2 C_2) \cos q_1 - a_2 S_2 \sin q_1$$
$$y = (a_1 + a_2 C_2) \sin q_1 + a_2 S_2 \cos q_1$$

Flash back from the Past

Sum and Difference Identities

- $\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$
- $\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$
- $\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$
- $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$
- $\tan(\alpha+\beta) = (\tan\alpha + \tan\beta)/(1 - \tan\alpha\tan\beta)$
  $\tan(\alpha-\beta) = (\tan\alpha - \tan\beta)/(1 + \tan\alpha\tan\beta)$

Analytic/Algebraic Solution

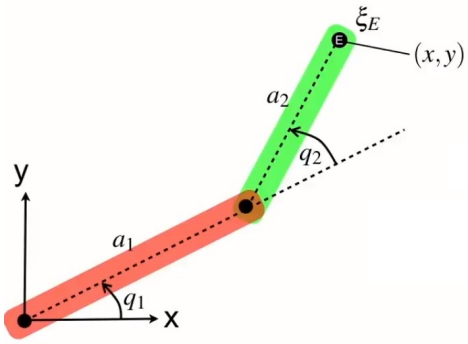


$$x = (a_1 + a_2C_2)\cos q_1 - a_2S_2\sin q_1$$
$$y = (a_1 + a_2C_2)\sin q_1 + a_2S_2\cos q_1$$

Famous Trigonometric Solution:

$$a\cos\theta + b\sin\theta = c$$
$$\theta = \tan^{-1}\frac{c}{\pm\sqrt{a^2 + b^2 - c^2}} - \tan^{-1}\frac{a}{b}$$

Analytic/Algebraic Solution



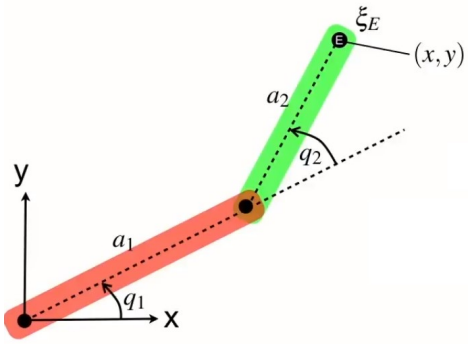
$$x = (a_1 + a_2C_2)\cos q_1 - a_2S_2 \sin q_1$$

$$y = (a_1 + a_2C_2)\sin q_1 + a_2S_2 \cos q_1$$

$$a\cos\theta + b\sin\theta = c$$

$$\theta = \tan^{-1}\frac{c}{\pm\sqrt{a^2+b^2-c^2}} - \tan^{-1}\frac{a}{b}$$

Analytic/Algebraic Solution



$$x = (a_1 + a_2C_2)\cos q_1 - a_2S_2 \sin q_1$$

$$y = (a_1 + a_2C_2)\sin q_1 + a_2S_2 \cos q_1$$

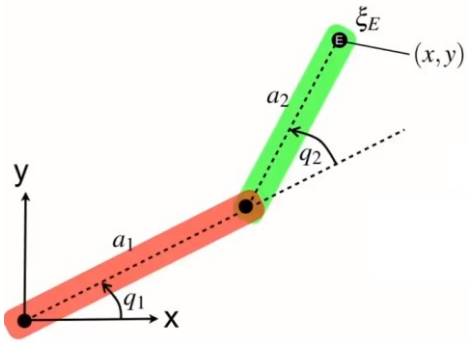
$$a\cos\theta + b\sin\theta = c$$

$$\theta = \tan^{-1}\frac{c}{\pm\sqrt{a^2+b^2-c^2}} - \tan^{-1}\frac{a}{b}$$

$$a = a_2S_2, b = a_1 + a_2C_2, c = y$$

Using a b and c write down the value of q1 on your notebooks now.

Analytic/Algebraic Solution



$$x = (a_1 + a_2C_2)\cos q_1 - a_2S_2 \sin q_1$$

$$y = (a_1 + a_2C_2)\sin q_1 + a_2S_2 \cos q_1$$

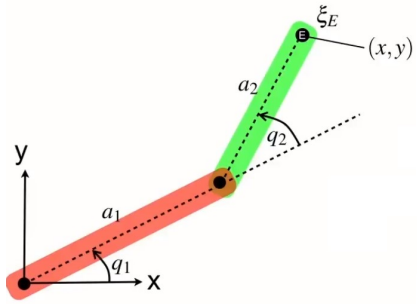
$$a\cos\theta + b\sin\theta = c$$

$$\theta = \tan^{-1}\frac{c}{\pm\sqrt{a^2+b^2-c^2}} - \tan^{-1}\frac{a}{b}$$

$$a = a_2S_2, b = a_1 + a_2C_2, c = y$$

$$q_1 = \tan^{-1}\frac{y}{\sqrt{a_1^2 + a_2^2 + 2a_1a_2C_2 - y^2}} - \tan^{-1}\frac{a_2S_2}{a_1 + a_2C_2}$$

Analytic/Algebraic Approach

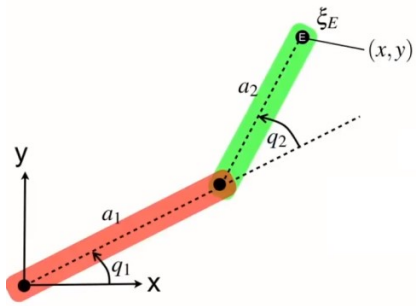


$$q_1 = \tan^{-1}\frac{y}{\sqrt{a_1^2 + a_2^2 + 2a_1a_2C_2 - y^2}} - \tan^{-1}\frac{a_2S_2}{a_1 + a_2C_2}$$

$$x^2 + y^2 = a_1^2 + a_2^2 + 2a_1a_2\cos q_2$$

Simplify this using the given equation which we found a little while ago.
Do on your notebook!

Analytic/Algebraic Approach



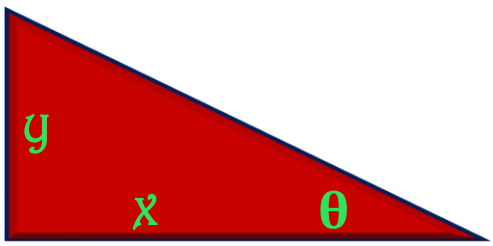
$$q_1 = \tan^{-1}\frac{y}{\sqrt{a_1^2 + a_2^2 + 2a_1a_2C_2 - y^2}} - \tan^{-1}\frac{a_2S_2}{a_1 + a_2C_2}$$

$$x^2 + y^2 = a_1^2 + a_2^2 + 2a_1a_2\cos q_2$$

$$q_1 = \tan^{-1}\frac{y}{x} - \tan^{-1}\frac{a_2S_2}{a_1 + a_2C_2}$$

Tangent Functions

- High school refresher



$$y = x\tan\theta$$

$$\tan\theta = \frac{y}{x}$$

$$\theta = \tan^{-1}\frac{y}{x}$$

$$\tan^{-1} \sim \text{atan} \sim \text{arctan}$$

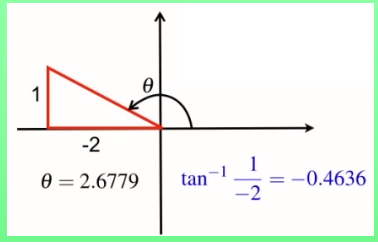
Range of tan inverse

$$\tan^{-1} x \in [-\pi/2, \pi/2]$$

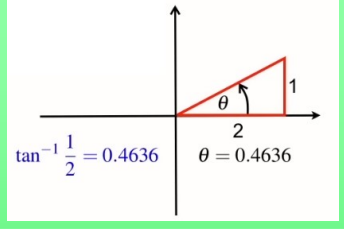
40

Consider Some Complex cases

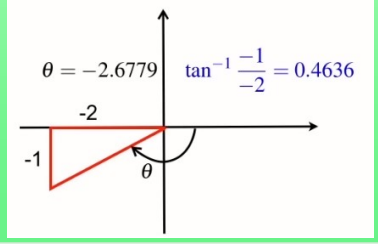
II



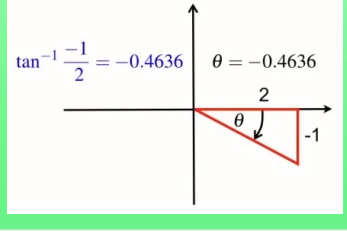
I



III



IV



41

Solution!

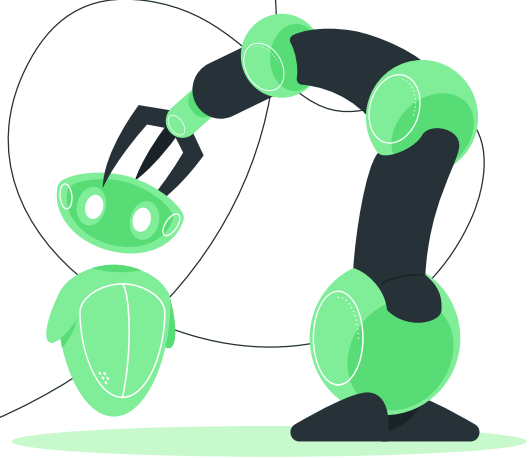
- To solve the sign ambiguity problem, we introduced a new function. You can find it in many programming libraries and of course MATLAB

$$\text{atan2}(a,b) = \tan^{-1}\frac{a}{b} \qquad \in [-\pi, \pi]$$

```
>> atan2(1,-2)
ans =
    2.6779
```


Inverse Kinematics for 3D robot

We'll do that in workshop



IK for a general Robot Arm in 3D

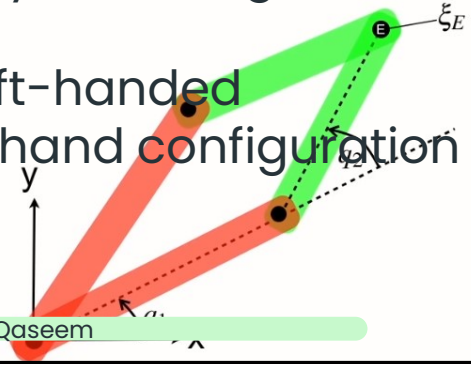
- Multiple possible configurations for the same pose
- Left/right-handed
- Elbow up/down
- Wrist position

Robot Arm Configuration change

- A characteristic of inverse kinematics is that there is often more than one solution, that is, more than one set of joint angles gives exactly the same end-effector pose.
- The end point remains the same whereas joint angles need to adjust themselves

Robot Arm Configuration change

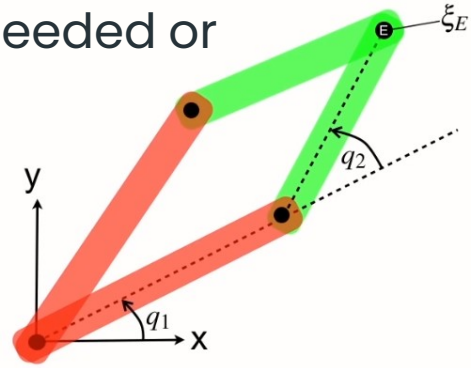
- How robot move from one configuration to other? Or more specifically can we compute the trajectory for configuration change?
- Example case from left-handed configuration to right hand configuration or vice versa.



46

Robot Arm Configuration change

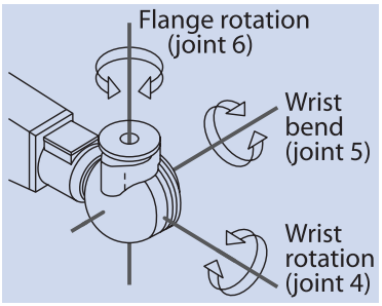
- Give one scenario where a change in Robot arm configuration from left to right or vice versa is needed or necessary.

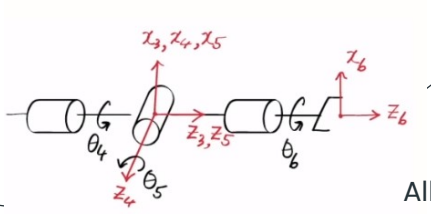


47

Spherical Wrist

All three wrist joints intersects at a single point.





All most all industrial robots have spherical wrists.

48

To sum up

- How many inverse kinematic solutions are there?

- Geometric
- Analytic/Algebric
- Numeric

49

Analytic Solution

- Wrote equations for forward kinematics
- Hard part: Inverse kinematics. Lots of equations to solve for joint variables
  - needs human insight
  - but can be automated for specific set/class of robot arm

Analytic Solution

- **Pros**
- Multiple robot configurations come out explicitly
- Solution expressions are compact and fast to execute
- **Cons**
- When the number of joint increases…?
- (search in literature)

Analytic Solution

- create a model of the robots joints and links
- write the forward kinematics
- write down lots of equations and attempt to solve for the unknown joint variables
  - ➥ needs some human insight
  - ➥ can be automated
- explicitly shows the multiple configurations
- solution are compact and fast to execute

- increasingly difficult as the number of joints increases
  - ➥ however many solutions have been derived and published

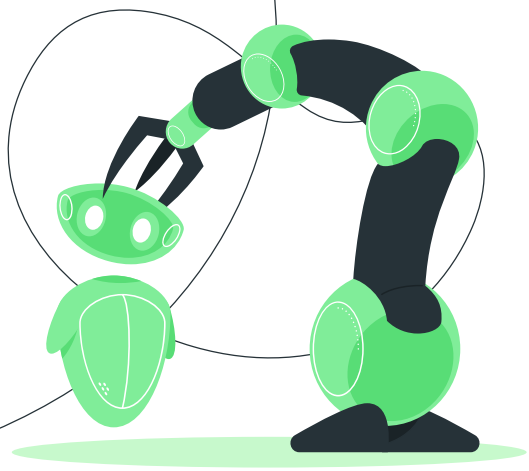
52

What if the Analytic Solution doesn't exist or is too hard?

53

We have another way! ☺

Inverse Kinematics Numerical Approach



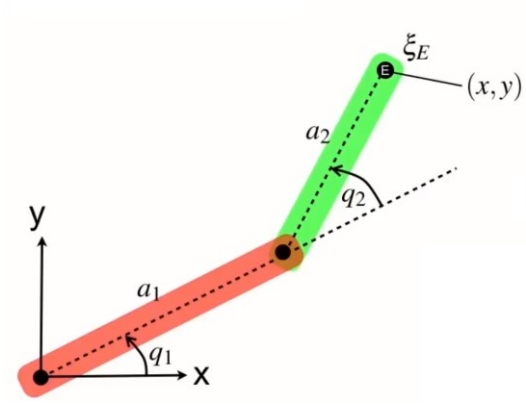
54

Numeric Solution

- We can always derive the forward kinematics of a Robot $\xi = \mathcal{K}(\boldsymbol{q})$
- In IK problem, we know the desired pose ξ^*
- To find q, we adjust q until $\xi \to \xi^*$
- This is formally achieved through optimization $\boldsymbol{q}^* = \arg\min_{\boldsymbol{q}} |\mathcal{K}(\boldsymbol{q}) \ominus \xi^*|$

55

IK through Numerical Analysis



$$\mathbf{E} = \mathbf{R}_x(q_1)\,\mathbf{T}_x(a_1)\,\mathbf{R}_x(q_2)\,\mathbf{T}_x(a_2)$$

$$\mathbf{E} = \begin{pmatrix} \cos(q_1+q_2) & -\sin(q_1+q_2) & a_2\cos(q_1+q_2)+a_1\cos q_1 \\ \sin(q_1+q_2) & \cos(q_1+q_2) & a_2\sin(q_1+q_2)+a_1\sin q_1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_2\cos(q_1+q_2)+a_1\cos q_1 \\ a_2\sin(q_1+q_2)+a_1\sin q_1 \end{pmatrix}$$

Activate Windows
Go to Settings to activate Windc

IK through Numerical Analysis

- **Problems?**
- Initial joint coordinates value is critical
- (NS may not converge if initial value is too far away)
- No knowledge/guarantee about a particular robot configuration
- Computationally expensive

Practical Issues

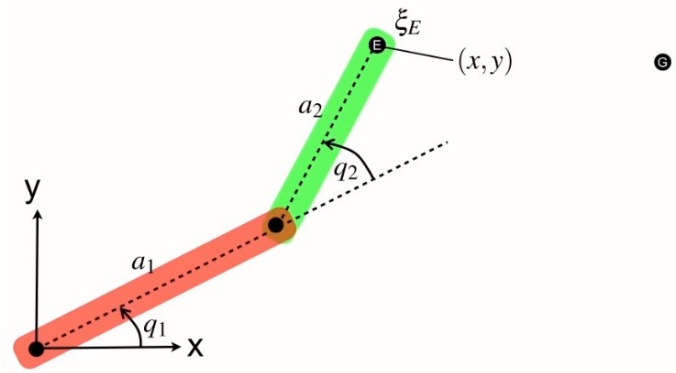
-



58

Practical Issues

- Reachability?



59

Practical Issues

- **Reachability?**
- Robot may not be able to reach a particular pose we want!
- That will cause problem in IK
- Numeric: Fail to converge
- Analytic: square root of –ve number
- Analytic: Inverse cosine of a number value > 1

60

Practical Issues

- **Singularity:**
- Loss of degree of freedom
- End-Effector is not capable of adopting particular orientations

61

Comparison

	Serial Link Robot	Parallel link Robot
FK	Easy, Analytic Solution exists	Complex, Analytic Solution may not exist
IK	Complex, Analytic Solution may not exist	Easy, Analytic Solution exists





62

Redundant Robots

-





63

Redundant Robots

- Robots with high degree of freedom

$$\mathcal{C} \subset \mathbb{R}^{100}$$

$$\mathcal{T} \subset \mathbb{SE}(3)$$

*A highly redundant robot with 100 joints has large configuration space

It is over actuated!

$$\xi_E \sim (x, y, z, \theta_r, \theta_p, \theta_y)$$

Pros?

Inverse Kinematics problem?

Quick Test!

64



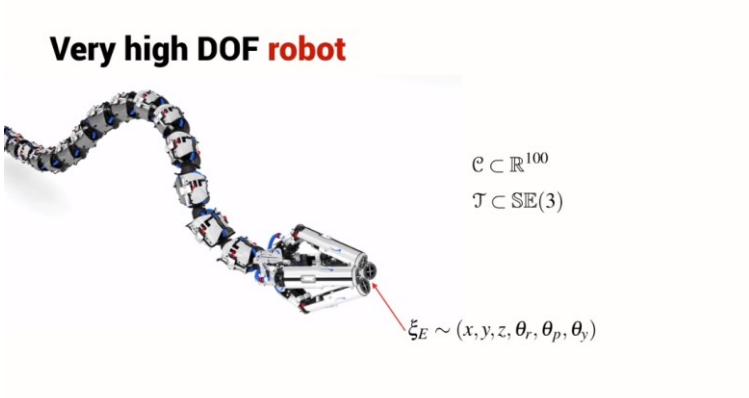
Quick Sleep Test

What is Configuration space of Puma Robot?

What is Task space of Puma Robot?

65

Redundant Robots



Which solution will you use for Inverse Kinematics? Numerical or Analytical?

Robotics – Umarah Qaseem

66

Credits

- This template is customized from a Slidesgo template.
- This lecture is made from the book “RVC by Professor Peter Corke”
- Some slides are inspired by Professor Peter Corke’s work.
- Rest of the material is self made and some of it is compiled from different internet resources.

slidesgo

Robotics – Umarah Qaseem

73